

Introduction to Quantum Computing

Part II Quantum Computing: Gates and Algorithms

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Chapter 12:

Quantum Register and Data Processing, Entanglement, the Bell States, and EPR Paradox

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Quantum Register

- In a classical computer, a register is a memory to store the classical bits of information.
- The size of the register determines the number of bits it can store. For example, a 4-bit register can store 4 bits of information, such as $(1001)_2$.
- The bit on the left is called the **Most Significant Bit** (MSB) and the bit on the right is called the **Least Significant Bit** (LSB).
- The MSB represents $1 \times 2^3 = 8$ and the least significant bit represents $1 \times 2^0 = 1$. The decimal value of this representation is $1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 9$.
- We can easily imagine how to implement an n -bit classical register physically even not knowing its physical implementation. Similarly, we can discuss and use **quantum register** in quantum algorithm without knowing how to implement a quantum register.
- Most of the time, a quantum register is only a grouping of qubits and it is not necessarily a memory as in the classical case. A quantum register is treated as a convenient way to group the qubits.
- A quantum register is just a natural extension of the concept of a classical register. An n -bit classical register stores n -bit of information while an n -qubit quantum register stores n -qubit of information.

Quantum Register

- For a 1-qubit register, it stores 1-qubit of information that represents a **vector** in the $\mathbb{C}^2$ spaces. We can write it as $|a\rangle = \alpha|0\rangle + \beta|1\rangle$, which is a linear combination of the basis vectors, $|0\rangle$ and $|1\rangle$.
- We see that when $|a\rangle = 0$ or $|a\rangle = 1$, the quantum register stores exactly the same information as the classical one.
- **If the quantum register is only allowed to store the basis states, it is the same as a classical register!**
- As mentioned, a quantum register is just a grouping of qubits, all the knowledge we learned earlier is directly applicable to the qubits stored in a quantum register. For a 4-qubit quantum register, it stores the information of 4 qubits, which is a vector in the $\mathbb{C}^{2^4} = \mathbb{C}^{16}$ space. This is obtained by performing the tensor products of 4 individual qubits, each from a $\mathbb{C}^2$ space.
- The vector or state, $|\psi\rangle$ that can be stored in a 4-qubit register is therefore a linear combination of the basis states.

Quantum Register

- Here we see the power of the quantum register. Instead of being able to store only the basis states (which is equivalent to the capability of a classical register), it can store the linear combination of the basis states. And the number of basis states, thus the dimension of the **Hilbert space**, grows exponentially with the number of qubits!
- Relationship between storable values in a 4-bit classical register and the basis states in a 4-qubit quantum register as following:

Value can be stored in a classical register	Basis states in a quantum register
$(0000)_2 = 0$	$ 0\rangle \otimes 0\rangle \otimes 0\rangle \otimes 0\rangle = 0\rangle 0\rangle 0\rangle 0\rangle = 0000\rangle = 0\rangle_{10}$
$(0001)_2 = 1$	$ 0\rangle \otimes 0\rangle \otimes 0\rangle \otimes 1\rangle = 0\rangle 0\rangle 0\rangle 1\rangle = 0001\rangle = 1\rangle_{10}$
$(0010)_2 = 2$	$ 0\rangle \otimes 0\rangle \otimes 1\rangle \otimes 0\rangle = 0\rangle 0\rangle 1\rangle 0\rangle = 0010\rangle = 2\rangle_{10}$
$\vdots$	$\vdots$
$(1111)_2 = 15$	$ 1\rangle \otimes 1\rangle \otimes 1\rangle \otimes 1\rangle = 1\rangle 1\rangle 1\rangle 1\rangle = 1111\rangle = 15\rangle_{10}$

- For an n -qubit register, it stores a vector in the 2^n dimensional space.

Quantum Register

- The basis states in the 2^n -dimensional vector space are orthonormal to each other. This is because for two basis states $|a\rangle = |a_{n-1}a_{n-2} \dots a_0\rangle$ and $|b\rangle = |b_{n-1}b_{n-2} \dots b_0\rangle$ where a_i and b_i can only take either “0” or “1”. So,

$$\langle a|b\rangle = \langle a_{n-1}a_{n-2} \dots a_0|b_{n-1}b_{n-2} \dots b_0\rangle = \langle a_{n-1}|b_{n-1}\rangle\langle a_{n-2}|b_{n-2}\rangle \dots \langle a_0|b_0\rangle$$

- $\langle a|b\rangle$ must be 0 if $a \neq b$ as they are different in at least one binary digit or 1 if $a = b$.
- A 2-qubit register stores a state in the $\mathbb{C}^4$ Hilbert space which is a tensor product space of two 1-qubit systems ($\mathbb{C}^2 \otimes \mathbb{C}^2$). A vector example in this space is,

$$|\psi\rangle = 0.5|0\rangle + 0.5|1\rangle + 0.5|2\rangle + 0.5|3\rangle$$

- In binary form they are $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$ which are the tensor products of the basis states of two $\mathbb{C}^2$ spaces, i.e. $|0\rangle \otimes |0\rangle$, $|0\rangle \otimes |1\rangle$, $|1\rangle \otimes |0\rangle$, and $|1\rangle \otimes |1\rangle$. The probability of finding the wavefunction to collapse to $|2\rangle$ is $0.5^2 = 0.25$.
- But what is the physical meaning of this? It is the same as finding the system to collapse to $|1\rangle \otimes |0\rangle$.

Quantum Data Processing

- Here is a fake example to show the power of quantum data processing.
- If we want to calculate a function, $f(x)$, for $x = 0, 1, 2, 3$ in a classical computer. We need to do it 4 times by calculating $f(0)$, $f(1)$, $f(2)$, and $f(3)$. We also need to store $x = 0, 1, 2, 3$ in 4 registers before calculation.
- In quantum computing, we just need to form a vector $|\psi\rangle$, which is a linear superposition of $|0\rangle$, $|1\rangle$, $|2\rangle$, and $|3\rangle$ and store it in one 2-qubit quantum register. And try to find a mechanism with a linear property so that we get

$$\begin{aligned} f(|\psi\rangle) &= f(0.5|0\rangle + 0.5|1\rangle + 0.5|2\rangle + 0.5|3\rangle) \\ &= 0.5f(|0\rangle) + 0.5f(|1\rangle) + 0.5f(|2\rangle) + 0.5f(|3\rangle) \end{aligned}$$

Then all the computations can be done at once!

- In summary, the power of quantum computing has two folds.
- Firstly, it only requires n -qubit to store a 2^n -dimensional space vector. If $n = 100$, we only need, for example, 100 electrons. But in a classical computer, to store a 2^{100} -dimensional space vector, we need to store its 2^{100} complex coefficients. This requires way more than the amount of classical memory we can imagine.
- Secondly, with the linearity of quantum mechanics, we can perform the calculation of the basis states simultaneously. If $n = 100$, we can perform 2^{100} calculations at one shot but this may take forever in a classical computer. This is the famous **quantum parallelism**. This is due to the linearity and **superposition** properties in quantum mechanics.

Quantum Data Processing

- In most important quantum algorithms, we utilize these two properties to perform gigantic calculations without the need for an enormous amount of hardware and by exploiting quantum parallelism.
- However, we should also realize that even we can store 2^n coefficients and perform 2^n computations easily in a quantum computer, it is difficult to read them due to wavefunction collapse after a single measurement.
- So, most quantum algorithms exploit the interference between the basis states to obtain the desired results.

Entanglement and Bell States

- Besides superposition and interference, entanglement is also a very heavily used concept and property in quantum computing algorithms.
- We know that every higher dimensional Hilbert spaces, e.g. $\mathcal{H}_3$, can be constructed by the tensor products of some lower-dimensional spaces, e.g. $\mathcal{H}_2$ and $\mathcal{H}_1$. For example, $\mathbb{C}^4 = \mathbb{C}^2 \otimes \mathbb{C}^2$.
- Any vector in the higher-dimensional space, e.g. $|h\rangle$, can be expressed as the linear combination of the tensor products of lower-dimensional space vectors. The equation is repeated here:

$$\begin{aligned} |h\rangle &= \sum_{j=1}^n |f_j\rangle \otimes |g_j\rangle \\ &= |f_1\rangle \otimes |g_1\rangle + |f_2\rangle \otimes |g_2\rangle + \cdots + |f_n\rangle \otimes |g_n\rangle \end{aligned}$$

- It is possible that for some higher-dimensional vectors, they can be expressed as just one single tensor product of two lower-dimensional vectors, each from a lower-dimensional space. That is

$$|h\rangle = |f\rangle \otimes |g\rangle$$

- In this case, n equals 1. These vectors are **unentangled**, which is not special. They are called **separable** or **product** states. However, there are some higher-dimensional vectors cannot be expressed as a single product of two lower dimensional vectors and must be expressed as the linear combination of them. Then we say the lower-dimensional vectors are **entangled**, which means that they cannot be separated.¹

Entanglement and Bell States

- Let's try an example in the $\mathbb{C}^4 = \mathbb{C}^2 \otimes \mathbb{C}^2$ space. Firstly, try to construct a vector in $\mathbb{C}^4$ using the tensor product of two vectors in the $\mathbb{C}^2$ spaces. Let $|f\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|g\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, we can construct

$$\begin{aligned}
 |h\rangle &= |f\rangle \otimes |g\rangle \\
 &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\
 &= \frac{1}{\sqrt{2}}(1 \times 1|0\rangle \otimes |0\rangle + 1 \times -1|0\rangle \otimes |1\rangle + 1 \times 1|1\rangle \otimes |0\rangle + 1 \times -1|1\rangle \otimes |1\rangle) \\
 &= \frac{1}{\sqrt{2}}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)
 \end{aligned}$$

Is the vector $|h\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$ entangled? No, because we just showed that it can be expressed as a single tensor product of two lower-dimensional vectors $|f\rangle$ and $|g\rangle$.

Entanglement and Bell States

- How about the vector $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$? This is also a vector in the $\mathbb{C}^4$ space. If we write out all basis, it is $|\phi^+\rangle = \frac{1}{\sqrt{2}}(1|00\rangle + 0|01\rangle + 0|10\rangle + 1|11\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$.

- Can we express it as a tensor product of two vectors from the $\mathbb{C}^2$ spaces? We will assume it can and $|\phi^+\rangle = |m\rangle \otimes |n\rangle$, where $|m\rangle = a_0|0\rangle + a_1|1\rangle$ and $|n\rangle = b_0|0\rangle + b_1|1\rangle$, then

$$\begin{aligned} |\phi^+\rangle &= |m\rangle \otimes |n\rangle \\ &= (a_0|0\rangle + a_1|1\rangle) \otimes (b_0|0\rangle + b_1|1\rangle) \\ &= a_0b_0|00\rangle + a_0b_1|01\rangle + a_1b_0|10\rangle + a_1b_1|11\rangle \end{aligned}$$

- But $|\phi^+\rangle = \frac{1}{\sqrt{2}}(1|00\rangle + 0|01\rangle + 0|10\rangle + 1|11\rangle)$, so the coefficients of the corresponding basis must be the same as those in above equation. Therefore

$$\begin{aligned} a_0b_0 &= 1 \\ a_0b_1 &= 0 \\ a_1b_0 &= 0 \\ a_1b_1 &= 1 \end{aligned}$$

- From the above equation, either a_0 or b_1 needs to be 0. Then it means either $a_0b_0 = 0$ or $a_1b_1 = 0$ which contradicts the first or the fourth equations. That means our assumption that $|\phi^+\rangle = |m\rangle \otimes |n\rangle$ is wrong and it is impossible to find $|m\rangle$ and $|n\rangle$ such that their tensor product is $|\phi^+\rangle$. Therefore, $|\phi^+\rangle$ is entangled.

Entanglement and Bell States

- $|\phi^+\rangle$ is special and it is also one of the four Bell states. In the $\mathbb{C}^4$ space, since it is 4-dimensional, it has four basis states. If we think of it as the space of two electron spins, it is natural to construct the basis to be $|\uparrow\uparrow\rangle$, $|\uparrow\downarrow\rangle$, $|\downarrow\uparrow\rangle$, and $|\downarrow\downarrow\rangle$ or in other words, $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$.
- We can also create a new basis using the so-called **Bell states**. The Bell states are all entangled
- and are very useful in certain applications. The Bells states are

$$|\phi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}; \quad |\phi^-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}$$

$$|\psi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}; \quad |\psi^-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

Entanglement and Bell States

- Let's explore a special property of an entangled state. Suppose we have two electron spins and they form an entangled state, ϕ^+ , in a system at time $t = 0$ year on the Earth. We then send one of the electrons to our alien friend 100 light-years away (i.e. it is so far that it takes the light 100 years to reach there) and we assume the state is not destroyed and the two electrons are still entangled. Let's say we are able to send the electron at half of the speed of the light. So after 200 years, at $t = 200$ year, our friend receives the electron. At $t = 201$ year, we decide to measure the spin of my electron by applying a magnetic field in the $\hat{z}$ direction and then measure its energy. What will we get?
- Of course, we will get either $|\uparrow\rangle$ or $|\downarrow\rangle$ because these are the two eigenstates of the system with the external magnetic field. And we know that after a measurement, it will randomly collapse to either $|\uparrow\rangle$ or $|\downarrow\rangle$. However, $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle)$. If our electron's wave function collapses to $|\uparrow\rangle$ or $|\downarrow\rangle$, the whole system must collapse to $|\uparrow\uparrow\rangle$ or $|\downarrow\downarrow\rangle$, respectively.
- This also means that our friend's electron wavefunction will collapse to the same spin state as our electron even our alien friend does not do the measurement and even it takes the light 100 years to tell the other electron what happened to our electron.
- This looks like we can transmit information instantaneously and at a speed faster than light. This is related to the Einstein–Podolsky–Rosen (EPR) paradox.

Einstein–Podolsky–Rosen (EPR) Paradox

- Albert Einstein who won his Nobel prize due to photon quantization (to explain the photoelectric effect) does not believe in quantum mechanics. He thinks God does not play dice. The randomness in quantum mechanics makes him very uncomfortable. With Podolsky and Rosen, they came up with a paradox to show that quantum mechanics is either wrong or incomplete.
- This is related to the entanglement experiment we just did with our alien friend.
- After we sent an electron in the entangled pair of $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle)$ to our alien friend in the hotel, we measure our electron in the $|0\rangle / |1\rangle$ basis. We assume we get $|0\rangle$. Then the state of the two-electron system becomes $|00\rangle$. We call our friend and say we got $|0\rangle$. Our friend then knows with a 100% certainty that the electron in the hotel must be $|0\rangle$ due to entanglement. Then, our friend decides to perform a measurement on the hotel electron in the $|+\rangle / |-\rangle$ basis. We know $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$. So the electron must collapse to either $|+\rangle$ or $|-\rangle$. Assume it collapses to $|+\rangle$. Now our friend also knows with a 100% certainty that the electron is in the $|+\rangle$ state. So our friend knows with 100% certainty that the electron is in the $|+\rangle$ and $|0\rangle$ states.
- But is not that due to the Heisenberg Uncertainty principle, we cannot measure $|+\rangle$ and $|0\rangle$ states with a 100% certainty as these two bases are incompatible. That means either the quantum mechanics is wrong or incomplete (with hidden variables we still have not discovered).

Einstein–Podolsky–Rosen (EPR) Paradox

- Einstein is thus not convinced that quantum mechanics is correct. There are experiments related to this (which is to prove the so-called Bell's Inequality) and they show that quantum mechanics “looks” correct.
- It only looks correct because there is a possibility that the experiment is not 100% well controlled. However, the technology has been improving that the chance that it is not well controlled becomes smaller and smaller which gives us more and more confidence that quantum mechanics is correct.

Chapter 13:

Concepts Review, Density Matrix, and Entanglement Entropy

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Concepts Review Using Entanglement

- Here, we will use an example of entanglement to review some of the most critical concepts and skills we have learned before moving to the next “steps.”
- The question is that if there is a Bell state on a certain basis, is it possible to express it on another basis so that it is not entangled? Or if it is entangled in a certain basis, is it always entangled no matter how you change the basis?
- For example, $|\psi^-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}$. This is a 4-dimensional space vector. We are using the basis formed by $|00\rangle/|01\rangle/|10\rangle/|11\rangle$ in the $\mathbb{C}^4$ Hilbert space, which are the tensor product of two $\mathbb{C}^2$ space with the basis vectors, $|0\rangle$ and $|1\rangle$. Therefore, we can also write

$$\begin{aligned} |\psi^-\rangle_{\{01\}} &= \frac{1}{\sqrt{2}} (0|00\rangle + 1|01\rangle - 1|10\rangle + 0|11\rangle) \\ &= \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle) \end{aligned}$$

- We already proved in the Problems in the last chapter that $|\psi^-\rangle$ is an entangled state and cannot be expressed as a single tensor product of two vectors from the $\mathbb{C}^2$ spaces. However, is it possible to have a Bell state on another basis (let's call it $\{\alpha\beta\}$), which is entangled, such that it is not entangled in the $\{01\}$ basis?

Concepts Review Using Entanglement

- $\mathbb{C}^2$ is a 2D space with complex coefficients. we can use a real 2D plane to illustrate how to change the basis as shown in Pic. 9. This picture is wrong but is intended to give us a relevant idea.
- In general, the new basis vectors $|\alpha\rangle$ and $|\beta\rangle$ can be expressed as the linear combinations (superposition) of the old basis vectors, $|0\rangle$ and $|1\rangle$:

$$|\alpha\rangle = a|0\rangle + b|1\rangle$$

$$|\beta\rangle = c|0\rangle + d|1\rangle$$

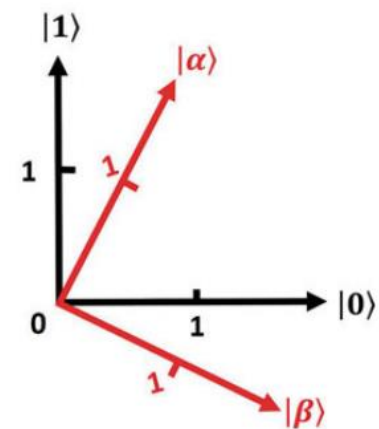
where a , b , c , and d are complex numbers. The basis vectors need to be orthonormal.

Therefore, $\langle\alpha|\alpha\rangle = \langle\beta|\beta\rangle = 1$ and $\langle\alpha|\beta\rangle = \langle\beta|\alpha\rangle = 0$. $|\beta\rangle$ has two complex numbers.

To determine $|\beta\rangle$, we need to fix 4 values (real and imaginary parts of the two complex numbers).

Equations $\langle\beta|\alpha\rangle = 0$ and $\langle\beta|\beta\rangle = 1$ help us fix two of them and it still has two numbers undetermined (i.e. still has two degrees of freedom).

One of them can be captured as the overall phase factor, which is immaterial because when we calculate any physical quantities, both the bra and ket versions of the vector are involved and the global phase factors in the bra and ket cancel each other.



Pic.9 Changing basis states in the $\mathbb{C}^2$ space using a fake real 2D space illustration

Concepts Review Using Entanglement

- Therefore, if $|\alpha\rangle$ is given, there is only one value left to be determined in $|\beta\rangle$. We may express $|\beta\rangle$ as

$$|\beta\rangle = e^{i\theta}(b^*|0\rangle - a^*|1\rangle) = \begin{pmatrix} e^{i\theta}b^* \\ -e^{i\theta}a^* \end{pmatrix}$$

let's check if it is normalized.

$$\begin{aligned} \langle\beta|\beta\rangle &= (e^{-i\theta}b - e^{-i\theta}a) \begin{pmatrix} e^{i\theta}b^* \\ -e^{i\theta}a^* \end{pmatrix} \\ &= e^{i\theta}be^{i\theta}b^* + (-e^{i\theta}a)(-e^{i\theta}a^*) = bb^* + aa^* = 1 \end{aligned}$$

Now let's check if $|\beta\rangle$ is orthogonal to $|\alpha\rangle$.

$$\begin{aligned} \langle\beta|\alpha\rangle &= (e^{-i\theta}b - e^{-i\theta}a) \begin{pmatrix} a \\ b \end{pmatrix} \\ &= e^{i\theta}ba + (-e^{i\theta}a)(b) = 0 \end{aligned}$$

Therefore, the equation $|\beta\rangle$ is a correct representation of $|\beta\rangle$ when $|\alpha\rangle$ is given and $|\beta\rangle$ and $|\alpha\rangle$ form a pair of orthonormal basis from which we will construct the new $|\phi^-\rangle$.

Concepts Review Using Entanglement

- Now we have two different bases in $\mathbb{C}^2$ space, namely, $\{01\}$ and $\{\alpha\beta\}$. They are related to each other. We can then construct a $\mathbb{C}^4$ space using either two $\{01\}$ spaces or two $\{\alpha\beta\}$ spaces. The new space will have four basis vectors. The straightforward sets are $|00\rangle/|01\rangle/|10\rangle/|11\rangle$ if it is formed by $\{01\}$ basis or $|\alpha\alpha\rangle/|\alpha\beta\rangle/|\beta\alpha\rangle/|\beta\beta\rangle$ if it is formed by $\{\alpha\beta\}$ basis.
- *Note that they are the same space but just have a different basis or they are “viewed” at different angles.*
- We represent them in column forms if we are using $|00\rangle/|01\rangle/|10\rangle/|11\rangle$,

$$|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}_{\{01\}} ; |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}_{\{01\}} ; |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}_{\{01\}} ; |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}_{\{01\}}$$

- How about in the $|\alpha\alpha\rangle/|\alpha\beta\rangle/|\beta\alpha\rangle/|\beta\beta\rangle$ basis? It is the same, except different basis label to avoid confusion.

$$|\alpha\alpha\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}_{\{\alpha\beta\}} ; |\alpha\beta\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}_{\{\alpha\beta\}} ; |\beta\alpha\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}_{\{\alpha\beta\}} ; |\beta\beta\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}_{\{\alpha\beta\}}$$

They are the same because each element in the column form represents how much the corresponding basis vector it has.

Concepts Review Using Entanglement

- Therefore, $|\psi^-\rangle_{\{\alpha\beta\}} = \frac{1}{\sqrt{2}}(|\alpha\beta\rangle - |\beta\alpha\rangle) = \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}_{\{\alpha\beta\}}$
- We want to know if this entangled $|\psi^-\rangle_{\{\alpha\beta\}}$ state is still entangled when it is represents in the $\{01\}$ basis.
- By using the distribution rule and making sure *not* to swap the orders of the ket's in the tensor products during this process, some terms are canceled, and we have

$$|\psi^-\rangle_{\{\alpha\beta\}} = -e^{i\theta} |\psi^-\rangle_{\{01\}}$$

Note that $-e^{i\theta}$ is just an overall phase factor and we can ignore it as discussed earlier. Therefore, the Bell state is still Bell state even after rotations or being represented on another basis. And, thus, it is still entangled.

Pure State, Mixed State, and Density Matrix

- In this class, we will only deal with **pure states**. However, to introduce the concept of entanglement measurement, it is necessary to discuss the **density matrix**, which inherently involves **mixed states**.
- A pure state is just a state (i.e. vector) in a space. For example, an electron spin of $|0\rangle$ is a pure state. But an electron spin of $\sqrt{0.7}|0\rangle + \sqrt{0.3}|1\rangle$ is also a pure state.
- A pure state is just a state in a space so it is not limited to the basis states. It can be any linear combination of the basis states. If we perform a measurement on the given pure state, there is a probability of 0.7 that we will get $|0\rangle$ and 0.3 that we will get $|1\rangle$, which are obtained by finding the square of the magnitude of the corresponding coefficient.
- However, it is often that we do not have complete information about the state. For example, there might be 100 electrons and we only know that 70% of them are $|0\rangle$ and 30% of them are $|1\rangle$. So each of the electron is in a pure state (either $|0\rangle$ or $|1\rangle$), but we do not know which one is at $|0\rangle$ or $|1\rangle$.
- Then we say the electrons are in mixed states. If we perform a measurement, there is still a probability of 0.7 that we will get $|0\rangle$ and 0.3 that we will get $|1\rangle$ just because we know that 70% of them are $|0\rangle$ and 30% of them are $|1\rangle$. Or in other words, we know for each electron, there is a 70% chance to be in $|0\rangle$ and 30% chance to be in $|1\rangle$.

Pure State, Mixed State, and Density Matrix

- We use **density matrix** to represent the **mixed state** mathematically. Density matrix of an ensemble (either mixed or pure) is defined as:

$$\rho = \sum_{j=0}^{n-1} p_j |\psi_j\rangle \langle \psi_j|$$

where we assume there are n different pure states, $|\psi_j\rangle$, each has the portion of p_j . Therefore, sum of p_j is 1 and it also represents the probability of finding an electron in state $|\psi_j\rangle$.

- For the system with pure state $|\psi\rangle = \sqrt{0.7}|0\rangle + \sqrt{0.3}|1\rangle$, it only has one pure state ($|\psi_0\rangle = |\psi\rangle$) and the density matrix is

$$\begin{aligned} \rho &= |\psi\rangle \langle \psi| = \begin{pmatrix} \sqrt{0.7} \\ \sqrt{0.3} \end{pmatrix} (\sqrt{0.7}\sqrt{0.3}) \\ &= \begin{pmatrix} 0.7 & \sqrt{0.21} \\ \sqrt{0.21} & 0.3 \end{pmatrix} \end{aligned}$$

Pure State, Mixed State, and Density Matrix

- For the system with the mixed state, we have

$$\begin{aligned} p &= 0.7|0\rangle\langle 0| + 0.3|1\rangle\langle 1| \\ &= 0.7\begin{pmatrix} 1 \\ 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \end{pmatrix} + 0.3\begin{pmatrix} 0 \\ 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \end{pmatrix} \\ &= 0.7\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + 0.3\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0.7 & 0 \\ 0 & 0.3 \end{pmatrix} \end{aligned}$$

where $|\psi_0\rangle = |0\rangle$, $|\psi_1\rangle = |1\rangle$, $p_0 = 0.7$, and $p_1 = 0.3$.

- We see that although the given pure and mixed states have the same probabilistic measurement outcomes, their density matrices are different.

Measurement of Entanglement

- To check if a pure state is an entangled state, we need to find its **entanglement entropy**. If it is 1, it is maximally entangled. If it is 0, it is a pure product state.
- Entanglement entropy is defined as

$$S_A = -T_{rB}(\rho_B \log_2 \rho_B) \text{ where } \rho_B = T_{rB}(\rho)$$

$T_r(\rho)$ is the trace of a matrix. The trace of a matrix is simply the sum of its diagonal elements. Therefore,

$$T_r(\rho) = \sum_{i=0}^{n-1} \rho_{i,i}$$

- For example, the trace of density matrix for the pure state $|\psi\rangle = \sqrt{0.7}|0\rangle + \sqrt{0.3}|1\rangle$ and the mixed state are both $\rho_{0,0} + \rho_{1,1} = 0.7 + 0.3 = 1$. But what is T_{rA} or T_{rB} ? This is called the partial trace in which the sum is only performed over the index of A or B subspaces, respectively. Since we are talking about entanglement. So, the space must be at least a tensor product of two subspaces (e.g. $\mathbb{C}^4 = \mathbb{C}^2 \otimes \mathbb{C}^2$).
- Let's consider $|\phi^-\rangle$. We know it is an entangled state. To what degree it is entangled?

Measurement of Entanglement

- $|\phi^-\rangle$ is a pure vector in the $\mathbb{C}^4$ space, which is a tensor product of two $\mathbb{C}^2$ spaces and let's call them space A and space B and thus $\mathbb{C}^4 = \mathbb{C}^2_A \otimes \mathbb{C}^2_B$.
- Firstly, evaluate the density matrix of $|\phi^-\rangle$.

$$\begin{aligned}\rho &= |\phi^-\rangle\langle\phi^-| = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix} \frac{1}{\sqrt{2}} (0 \ 1 \ -1 \ 0) \\ &= \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}\end{aligned}$$

Now let's compute $\rho_B = T_{rA}(\rho)$, it is just $0 + 1 + 1 + 0 = 2$ by tracing over both the first and the second spaces (i.e. A and B). However, if we only want to trace over space A , which is the first space in the tensor product, we need to keep space B indices intact. Pic. 10 shows the methodology. Like regular tracing, we only care about the elements with row index of space A equal to the column index of space A . Since we are only tracing A , there are 2 blocks and 8 elements that meet this criterion.

Pic.10 Illustrations on how to do a partial tracing. Left: tracing only the first qubit. Right: tracing only the second qubit. Those with the same colors will be added during tracing

Measurement of Entanglement

- The following shows the indices and the corresponding values.

$$\frac{1}{2} \begin{pmatrix} \rho_{\{00\},\{00\}} & \rho_{\{00\},\{01\}} & \rho_{\{00\},\{10\}} & \rho_{\{00\},\{11\}} \\ \rho_{\{01\},\{00\}} & \rho_{\{01\},\{01\}} & \rho_{\{01\},\{10\}} & \rho_{\{01\},\{11\}} \\ \rho_{\{10\},\{00\}} & \rho_{\{10\},\{01\}} & \rho_{\{10\},\{10\}} & \rho_{\{10\},\{11\}} \\ \rho_{\{11\},\{00\}} & \rho_{\{11\},\{01\}} & \rho_{\{11\},\{10\}} & \rho_{\{11\},\{11\}} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 0 & & \\ 0 & 1 & & \\ & & 1 & 0 \\ & & 0 & 0 \end{pmatrix}$$

- Since we need to keep the B index intact during the partial trace calculation, we will add the elements with the same B index and therefore, the partial trace over A is a 2×2 matrix.
- For example, $\rho_{\{00\},\{00\}}$ is added to $\rho_{\{10\},\{10\}}$ because it has the same B index (i.e. 00). Therefore,

$$\rho_B = \begin{pmatrix} 0 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} + \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$$

where ρ_B is the **reduced density matrix** of space B because its element still depends on the B indices.

Measurement of Entanglement

- We can then calculate the entanglement entropy accordingly.

$$\begin{aligned}
 S_A &= -T_{rB}(\rho_B \log_2 \rho_B) \\
 &= -T_{rB} \left(\rho_B \log_2 \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \right) \\
 &= -T_{rB} \left(\rho_B \begin{pmatrix} \log_2 \frac{1}{2} & 0 \\ 0 & \log_2 \frac{1}{2} \end{pmatrix} \right) \\
 &= -T_{rB} \left(\rho_B \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \right) \\
 &= -T_{rB} \left(\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \right) = -T_{rB} \begin{pmatrix} -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} = 1
 \end{aligned}$$

- Therefore the entanglement entropy of $|\phi^-\rangle$ is 1, which means that the entanglement is maximal.
- Since the trace of a matrix does not change when the basis is changed, we can just work in the basis formed by the eigenvectors of ρ_B .

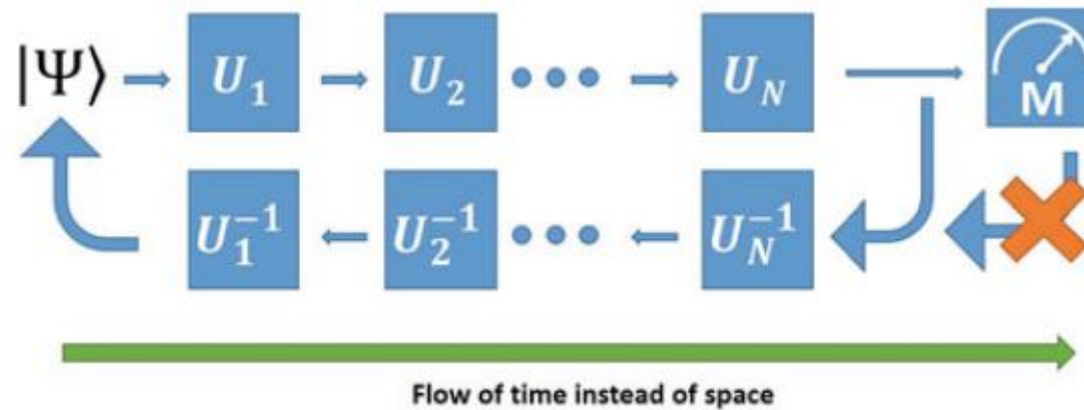
Chapter 14:

Quantum Gate Introduction: NOT and CNOT Gates

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Basic Quantum Gate Properties

- What is quantum computing? In short, it is just a series of rotations of a high *dimensional vector* in a *high-dimensional space*. This high-dimensional space is the tensor product of many lower dimension spaces. For what we are interested in here, the basic low-dimensional space is the $\mathbb{C}^2$ Hilbert space where the qubit vectors reside.
- For example, in a 4-qubit quantum algorithm, the vector of interest is a 16-dimensional vector in the $\mathbb{C}^{16}$ space, which may be constructed by grouping four electron spins together.
- At the end of the rotations, we then perform a measurement to obtain the answer we are looking for as Pic. 11.



Pic.11 Overview of quantum computing. U_1 and U_1^{-1} , etc. are quantum gates, which are unitary

Basic Quantum Gate Properties

- We need to perform meaningful rotations, so that we obtain the result we want. Besides that, the rotation needs to be unitary as we have discussed before.
- A unitary matrix, U , obeys $U^{-1} = U^\dagger$. The most important property of a unitary matrix is that it preserves the inner product of vectors and this is required in the laws of physics. Since it is unitary, it means its inverse exists.
- This is another requirement of the quantum gate. This is because quantum mechanics requires that every evolution (rotation) of the wavefunction is **reversible**. This is natural if we think about the operations as the rotations in a hyperspace. For a rotation, it is always possible to rotate it backwards if we are given the result (the output) and the nature of the rotation (the type of the gate).
- Every step in quantum computing is unitary and reversible. However, at the measurement, this is no longer true.
- A quantum gate is a gate that performs a unitary operation. In a classical computer, in general, the electrical signals flow through some physical gates (e.g. a NAND gate formed by 4 transistors) in the space. In quantum computing, usually, the qubit and its “signal” do not flow in the space. Instead, we apply operations such as microwave or laser pulses to operate/manipulate/change the qubit.
- Therefore, the flow of a quantum algorithm usually represents the flow of time instead of a physical layout in the space.

Basic Quantum Gate Properties

- The qubits are usually stationary (except that in a trapped-ion quantum computer, the trapped ions are moved around but this still does not completely resemble the flow of signals in a classical computer).
- When the external pulses or fields are applied to a physical qubit, this is just physics and we know that it creates a system in which the qubit state will evolve accordingly. In physics, how the state changes depends on the total energy landscape of the system. The total energy is called the **Hamiltonian**.
- The law that describes how the state in the system evolves is called the **Schrödinger equation**, which is equivalent to Newton's Laws in classical mechanics. Schrödinger equation is written as

$$i\hbar \frac{\partial |\psi\rangle}{\partial t} = \mathbf{H}|\psi\rangle$$

where $\hbar$ is the reduced Planck's constant, $\mathbf{H}$ is the Hamiltonian, and $|\psi\rangle$ is the state (vector) of the system. The left-hand side of the equation is the rate of change of the state. The right-hand side tells us that the rate of the change of the state can be obtained by multiplying the Hamiltonian on the state. This is as simple as Newton's $F = ma$!

- $|\psi\rangle$ is a vector and can be expressed in a column form. $\mathbf{H}$ is an operator and, thus, is a matrix.
- From the Schrödinger equation, the solution is: $|\psi(t)\rangle = e^{\frac{-i\mathbf{H}t}{\hbar}}|\psi(0)\rangle$. Therefore, the wavefunction (or the state vector) at time t , $|\psi(t)\rangle$, is just equal to $e^{\frac{-i\mathbf{H}t}{\hbar}}$ multiplied by the initial wavefunction at time 0, $|\psi(0)\rangle$.

Basic Quantum Gate Properties

- $|\psi(t)\rangle$, and $|\psi(0)\rangle$ are vectors and, thus, $e^{\frac{-iHt}{\hbar}}$ must be a matrix (it can also be a scalar, but it is not in this case). How do we compute $e^{\frac{-iHt}{\hbar}}$? In other words, how do we exponentiate a matrix?
- If the matrix is diagonal, we just need to exponentiate the diagonal elements. If not, we need to diagonalize it first.
- Let's assume we are working in the basis formed by the eigenvectors of $\mathbf{H}$ so that $\mathbf{H}$ is diagonal. By substituting as following:

$$\begin{aligned} i\hbar \frac{\partial e^{\frac{-iHt}{\hbar}} |\psi(0)\rangle}{\partial t} &= \mathbf{H} e^{\frac{-iHt}{\hbar}} |\psi(0)\rangle \\ i\hbar \frac{-iH}{\hbar} e^{\frac{-iHt}{\hbar}} |\psi(0)\rangle &= \mathbf{H} e^{\frac{-iHt}{\hbar}} |\psi(0)\rangle \\ \mathbf{H} e^{\frac{-iHt}{\hbar}} |\psi(0)\rangle &= \mathbf{H} e^{\frac{-iHt}{\hbar}} |\psi(0)\rangle \end{aligned}$$

Therefore, $e^{\frac{-iHt}{\hbar}}$ is a general expression of any quantum gate and let us call it $\mathbf{U}$. Let's check if it is unitary. Before that we need to recognize that $\mathbf{H}$, which is the operator for energy, is Hermitian because energy is an observable. Therefore, $\mathbf{H} = \mathbf{H}^\dagger$. We have

$$\begin{aligned} \mathbf{U}\mathbf{U}^\dagger &= e^{\frac{-iHt}{\hbar}} \left(e^{\frac{-iHt}{\hbar}} \right)^\dagger \\ &= e^{\frac{-iHt}{\hbar}} e^{\frac{iH^\dagger t}{\hbar}} = e^{\frac{-iHt}{\hbar}} e^{\frac{iHt}{\hbar}} = \mathbf{I} \end{aligned}$$

NOT (X) Gate

- The NOT gate is called X gate. Since every quantum gate is a unitary gate, we will label its matrix as U_{NOT} . It is called the NOT gate because it is defined in this way:

$$U_{NOT}|0\rangle = |1\rangle$$

$$U_{NOT}|1\rangle = |0\rangle$$

- It is exactly the same as the classical NOT gate if we just treat the basis states as the classical values.
- To achieve the above function, we found that

$$U_{NOT} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Since a qubit state is in the two-dimensional $\mathbb{C}^2$ space, the matrix (operator) must be 2×2 .

$$\begin{aligned} U_{NOT}|0\rangle &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} |0\rangle \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle \end{aligned}$$

and

$$\begin{aligned} U_{NOT}|1\rangle &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} |1\rangle \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle \end{aligned}$$

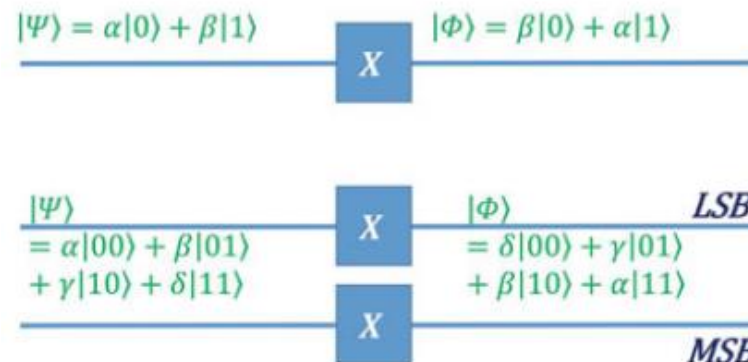
Circuit and Properties

- Just like the power of a quantum register is that it can store the superpositions of the basis states, the power of the quantum gates is that they can process the superpositions of the basis states. If U_{NOT} is applied to a general state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.

$$\begin{aligned} U_{NOT}|\psi\rangle &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} |\psi\rangle \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \beta \\ \alpha \end{pmatrix} = \beta|0\rangle + \alpha|1\rangle \end{aligned}$$

Therefore, we can also understand U_{NOT} as a gate that will swap the coefficients of the basis states (i.e. α becomes β and vice versa).

- The NOT gate is also called an X gate because the matrix U_{NOT} is just σ_x . Pic. 12 shows the NOT operation and its circuits.



Pic.12 NOT gate and its circuits. Top: 1-qubit NOT gate.
Bottom: 2-qubit NOT gate. Note the bottom qubit is the MSB

Circuit and Properties

- We can also construct a two-qubit or n-qubit NOT gate. We just need to perform a tensor product of two one-qubit NOT gate to obtain a two-qubit NOT gate because a two-qubit vector resides in a $\mathbb{C}^4$ space, which is a tensor product of two $\mathbb{C}^2$ spaces. Therefore,

$$\begin{aligned} U_{NOT2} &= U_{NOT} \otimes U_{NOT} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ 1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 0 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

- When apply the 2-qubit NOT gate to a general vector $|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$.

$$U_{NOT2}|\psi\rangle = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix} = \begin{pmatrix} \delta \\ \gamma \\ \beta \\ \alpha \end{pmatrix}$$

- Note that we will follow IBM-Q qiskit's convention to assign the lower bit in the circuit diagram to be the most significant bit (MSB). The MSB means that qubit is written on the left in a tensor product form. For example, the qubit $a_0|0\rangle + b_0|1\rangle$ is the MSB in $|\psi\rangle = (a_0|0\rangle + b_0|1\rangle) \otimes (a_1|0\rangle + b_1|1\rangle)$, while $a_1|0\rangle + b_1|1\rangle$ is the least significant bit (LSB).

XOR (CNOT) Gate

- In classical logic, the XOR gate is a 2-input gate. Similarly, an XOR quantum gate is 2-qubit. It is more commonly known as the CNOT gate, which stands for Controlled-NOT gate for an obvious reason to be explained soon. An XOR quantum gate is defined as

$$U_{XOR}|ab\rangle = |aa \oplus b\rangle$$

- $|ab\rangle$ is a vector in the $\mathbb{C}^4$ space and a and b are either 0 or 1. $|ab\rangle$ can be any of the basis states, namely, $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$.
- After the XOR gate is applied to one of the basis states, it will become another basis state $|aa \oplus b\rangle$. $|aa \oplus b\rangle$ is just $|a, a \oplus b\rangle$ or becomes $a \oplus b$, where $\oplus$ is the classical *exclusive or* (XOR) logical operation. Let's see how it changes the individual basis state.

$$U_{XOR}|00\rangle = |0, 0 \oplus 0\rangle = |0, 0\rangle = |00\rangle$$

$$U_{XOR}|01\rangle = |0, 0 \oplus 1\rangle = |0, 1\rangle = |01\rangle$$

$$U_{XOR}|10\rangle = |1, 1 \oplus 0\rangle = |1, 1\rangle = |11\rangle$$

$$U_{XOR}|11\rangle = |1, 1 \oplus 1\rangle = |1, 0\rangle = |10\rangle$$

We see that U_{XOR} swaps the third and the fourth basis and keeps the rest unchanged.

Matrix for XOR (CNOT) Gate

- The matrix of U_{XOR} is

$$U_{XOR} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

since this is a 2-qubit gate for operating on a vector in the four-dimensional $\mathbb{C}^4$ space, the matrix (operator) must be 4×4 . Moreover, it only swaps the third and the fourth basis states. Therefore, this is an appropriate construction. Let us try a numerical example:

$$U_{XOR}|10\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = |11\rangle$$

This is because $|10\rangle$ is the third basis state and thus it is $\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$ and $|11\rangle$ is $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$.

Circuit and Properties

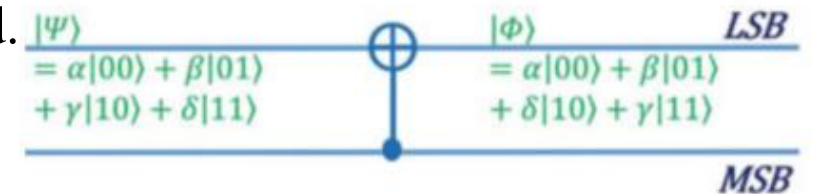
- The output depends on $a \oplus b$, where a is the value of the first qubit and b is the value of the second qubit in the basis state. It is also called Controlled-NOT. Whether we will apply NOT to the second qubit depends on a (i.e. controlled by a).
- The first qubit is called the control qubit and the second one is called the target qubit.
- In general, for a 2-qubit vector, $|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$, we have

$$U_{XOR}|\psi\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \\ \delta \\ \gamma \end{pmatrix}$$

$$= \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

As expected, the coefficients of the last two basis states are swapped.

Pic. 13 shows the circuit of a CNOT operation. Note that again the bottom qubit is the MSB and that qubit acts as the control qubit.



Pic.13 CNOT gate and its circuit